

# When Should a Streaming Model Speak?

## Per-Task Gate Thresholds for Qwen2.5-Omni on OmniPro Online

**Abstract.** A proactive streaming assistant must decide, at every moment, whether to speak, and that decision is almost always a scalar confidence compared against a per-task threshold. We fit such a threshold table for Qwen2.5-Omni-7B on OmniPro Online and report that it cannot be identified from the available data — together with the measurements that establish why, and the one change to the same gate that does work. Auditing first, we find the shipped gate ignored its own threshold table (firing on a fixed global constant) and treated the first tick of every video as a rising edge, consuming an entire video’s emission budget before any content arrived; both are repaired before measurement. We then test each fitted threshold by a paired bootstrap over videos that replays the actual selection rule. No task’s selection beats its best rival outside noise, selections recur in only 25–61 % of resamples, and one recurs at 6 % against an 8 % chance rate. The cause is that the gating score does not rank event-adjacent ticks above quiet ones: per-task AUC is 0.431–0.551 with every interval covering chance, and this survives sweeping the tick clock over  $\pm 10$ s, widening the matching tolerance, and confirming the score is well spread over 232–284 distinct values. A threshold on such a score buys emission volume only — and volume has a single useful setting, which pooled over all tasks is recovered stably in 94 % of resamples while nine per-task values are not recovered at all. We argue that a per-task threshold table is a set of fitted parameters and should be held to the corresponding standard: a resampling check that the values recur, and a comparison against one global threshold. Both re-use existing runs at no additional compute cost, and here each independently reverses the conclusion the point estimates would have supported.

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## 1 Introduction

A proactive streaming assistant watches a video as it arrives and must decide, at every moment, whether to speak. Almost universally that decision is implemented as a scalar confidence compared against a threshold, and almost as universally the threshold is tuned per task, because different tasks plainly want different eagerness: an alert that fires once per video is not the same object as a running commentary. Tuning nine thresholds for nine tasks and reporting the resulting gain is a standard move.

This paper performs that fit carefully for Qwen2.5-Omni-7B on OmniPro Online, and reports that it does not work — along with the measurements that establish *why* it does not, and the one intervention on the same gate that does.

We make three contributions.

*An audit that the fitted quantity is the used quantity.* Before fitting, we found that the shipped gate ignored the parameter we intended to fit: its fire test was a fixed global 0.5, and the per-task threshold table, the gate mode and the refractory period were inert. We also found that  $p_{\text{hit}}$  is high on the first tick of every video — the model answers the gating question from the prompt before it has seen any of the video — which under an edge-triggered gate with a long refractory consumed an entire video’s emission budget at  $t = 1.0\text{s}$ . Both are repaired before any number in this paper is measured. We report them because a fit of an inert parameter is the failure mode that a well-run grid search is least able to detect on its own.

*A test of whether a fitted threshold is a measurement.* A grid of ten cells scored on fifteen videos always has a maximum. We test each selection by a paired bootstrap over videos that replays the actual selection rule, and report a re-selection rate against chance alongside a confidence interval on the difference from the best rival. Under this test no task’s fitted threshold beats its own best alternative outside noise, and the fits re-select themselves 25–61 % of the time — one of them at 6 %, below its 8 % chance rate. Two apparent fits dissolve entirely on inspection: one task scores exactly 0.000 at all ten thresholds, and another’s entire threshold curve is one matched event, or zero, out of sixteen.

*A mechanism, and a positive result.* The reason is measurable rather than conjectural:  $p_{\text{hit}}$  does not rank event-adjacent ticks above quiet ones. Per-task AUC is 0.431–0.551 with every 95 % interval containing 0.5, and this survives a  $\pm 10\text{s}$  sweep of the tick clock, a widened  $\pm 10\text{s}$  tolerance, and a check that the score is not degenerate (232–284 distinct values spanning 0.000–0.997). A threshold applied to a score that cannot say *when* can only buy emission *volume* — and volume has exactly one useful setting. Pooled over 135 videos the single global operating point is identified and stable, re-selected in 94 % of resamples, and moving it from the shipped gate’s effective 0.5 is the one change here with a reproducible effect. Split nine ways over fifteen videos each, the same quantity is not identifiable at all.

The result is therefore not that thresholds do not matter. It is that this gate has one identifiable degree of freedom rather than nine, and that the difference between those two statements is invisible to any protocol that reports an argmax without asking whether it recurs.

## 2 Related work

*Proactive and streaming video dialogue.* A line of recent systems asks a model not merely to answer questions about a video but to decide *when* to speak while

the video is still arriving. MMDuet, LiveStar, MiniCPM-o and related streaming assistants each couple a perception backbone to some emission policy, and each reports aggregate scores on its own evaluation subset. OmniPro Online supplies a common 2,700-sample benchmark with per-event timing ground truth, which is what makes a timing metric such as time-F1 well defined.

We deliberately do not present cross-system numbers as a leaderboard. Published figures for these systems are computed on different subsets with different matching tolerances and different judges; quoting them beside ours in a single column would imply a comparability that does not exist. Where they appear, they are footnoted with the subset and scoring they came from.

*When to speak, as a threshold.* The dominant treatment of the emission decision is a scalar confidence compared against a threshold, frequently tuned per task or per dataset. Reported gains from such tuning are usually presented as a single number per configuration on the data the tuning was performed on. Our contribution is orthogonal to any particular policy: we ask what evidence is required before a per-task threshold table can be called a fitted parameter rather than a re-description of the sample it was fitted on, and we apply that test to a fit of our own.

*Confidence calibration versus ranking.* The distinction this paper turns on is standard in the calibration literature but rarely applied to streaming emission gates: a score can be well spread and still carry no information about the quantity being thresholded. Threshold selection can only exploit *ranking* — whether event-adjacent moments score above quiet ones — and ranking is measured by AUC, which is threshold-free and therefore unaffected by any fitting we do. We report AUC alongside F1 for exactly this reason: AUC measures whether the perception is right, F1 measures whether the gate is tuned, and two systems can be indistinguishable in the second while separable in the first.

### 3 Method

#### 3.1 The streaming system

The system is a three-thread proactive pipeline over a single shared KV cache. An *inger* decodes video at 1 fps and audio in 2 s chunks and appends both to the cache. A *controller* wakes on every tick, and a *writer* decodes an utterance whenever the controller decides to speak. The backbone is Qwen2.5-Omni-7B; the same harness previously ran a vision-only Qwen3-VL-8B, and the pipeline is unchanged apart from the backbone and the audio path.

At each tick the controller poses a boolean question to the model and reads  $p_{\text{hit}} = P(\text{true})$  directly off the logits at the answer slot. This costs one forward pass and *zero* decode steps, which is what makes a per-tick gate affordable: the expensive answer decode happens only when the gate fires. The consequence for cost is that a low threshold is substantially more expensive than a high one, because it admits more decodes — measured at  $\text{xRT} = 2.72$  at threshold 0.05 against a quiet-path cost of 0.3 s per tick with zero generated tokens.

### 3.2 The gate is three coupled knobs, not one

The decision to speak is governed by three parameters that cannot be fitted independently:

- **hit\_threshold** — the level  $p_{\text{hit}}$  must reach.
- **gate\_mode** — *level* (fire whenever above) or *edge* (fire only on a rising crossing).
- **refractory\_s** — the minimum silence after an emission.

The coupling is not incidental. Under *edge* mode with a long refractory, a single spurious early crossing can consume a video’s entire emission budget, so the cost of a mis-set threshold scales with the refractory. A threshold reported without its mode and refractory does not reproduce.

### 3.3 Two defects in the shipped gate, and their repair

Auditing the shipped configuration before fitting anything revealed that the quantity we intended to fit was not the quantity the system used.

*The threshold was inert.* Under the shipped `gate_strategy="hysteresis"`, the fire test was  $p_{\text{hit}} \geq \text{gate\_high\_thr}$ , a fixed global 0.5; `hit_threshold` only decided whether an answer was subsequently decoded, and `gate_mode` and `refractory_s` were read nowhere at all. Across a completed 2,700-sample run (581,403 ticks, 50,433 fires) not one fire occurred below  $p_{\text{hit}} = 0.500$ . Every task configured at or under 0.5 was therefore running at 0.5. We add a "fitted" strategy in which all three knobs are live, and report the shipped behaviour honestly as *a fixed global 0.5* rather than as the per-task table that was nominally configured.

*The first tick was a spurious rising edge.*  $p_{\text{hit}}$  is high on the first tick of every video: the model answers the boolean from the prompt before it has seen anything ( $\geq 0.05$  for 98–100 % of samples in every task, median 0.34–0.64). Seeding the edge detector with `prev_above = FALSE` made that tick a rising edge. For a task with a long refractory this consumed the video’s only emission: every `instant_event_alert` sample emitted once, at 1.0s, against ground truth at 26–98s. The fix is definitional rather than a tuned guard — an edge requires a predecessor, so `prev_above` starts undefined and the first tick can never be a crossing. Offline best time-F1 rose  $0.0535 \rightarrow 0.0900$  on `snapshot_counting` and  $0.0636 \rightarrow 0.0751$  on `instant_event_alert`, while short-refractory tasks moved by  $\pm 0.001$ . That the size of the effect tracks the refractory is the practical demonstration that the three knobs are one object.

## 4 Protocol

### 4.1 Metrics, and what we refuse to report

**time-F1** is judge-free: emissions are matched to ground-truth events by a greedy one-to-one assignment within  $\pm 3$ s, and precision, recall and F1 are computed from the pooled tp/fp/fn counts. All pooling in this paper is *micro-averaged* over samples, never the mean of per-sample ratios; the two differ, and only the former is comparable across tasks with different event densities.

**content-acc** and **joint-F1** require an LLM judge. When the judge is unreachable these are reported as WITHHELD, never estimated, never replaced by a dash that a reader would parse as zero. A lower bound that assumes every unjudged match is wrong is reported separately and under a name that says what it is; it is not a substitute for the measurement.

Every F1 in this paper travels with  $n$ ,  $n_{gt}$  and  $n_{emit}$ . A bare F1 on a 15-video cell is not interpretable, and several results below turn on exactly that point.

### 4.2 The fit subset and the two-pass grid

Thresholds are fitted on a frozen 5% subset: 135 videos, 15 per task, drawn all-audio and fixed before any fitting. Pass 1 sweeps a coarse grid of 10–12 thresholds per task (0.05...0.95, extended to 0.01 and 0.02 for the two tasks whose shipped values sit near zero); pass 2 refines around the pass-1 choice. Selection ranks by time-F1 — a deviation from the pre-registered joint-F1, forced by the judge being unreachable and recorded as such — with a 0.02 tie-band that prefers the emission volume closest to the truth when the top two are within it, so a winner is not manufactured out of emission count alone.

### 4.3 The noise floor, and why it governs every claim

The evaluation is not bit-reproducible: a fixed configuration re-run end to end produced time-F1 0.255 against 0.051 on one task. We therefore treat differences below 0.03 in pooled time-F1 as noise, and we say so before reporting any comparison rather than after seeing which way it fell.

A 15-video cell holds roughly 36 scoring events at the benchmark’s 2.4 events per sample. A gap of 0.02–0.03 between adjacent thresholds is not a real difference at that  $n$ , so a task whose curve is flat or non-monotonic across the whole grid is reported as FLAT and *keeps its shipped value* rather than being assigned a fitted one.

### 4.4 Reproducibility of a maximum

The central methodological commitment of this paper is that reporting the argmax of a grid is not the same as reporting a fitted parameter. A grid of ten cells scored on fifteen videos always has a maximum. Whether that maximum is a measurement depends on whether it recurs.

**Table 1.** Finalized per-task gate thresholds.  $n$ ,  $n_{gt}$  and  $n_{emit}$  are reported beside every F1: at  $\sim 2.4$  GT events per sample a 15-sample cell holds  $\sim 36$  scoring events, and a 0.02–0.03 gap between adjacent thresholds is not a real difference.

| task            | ship  | p1    | 2nd   | final        | $\Delta$ | time-F1 | joint-F1 | $n$ | e/gt |
|-----------------|-------|-------|-------|--------------|----------|---------|----------|-----|------|
| instant alert   | 0.450 | 0.050 | 0.150 | <b>0.450</b> | +0.000   | 0.0000  | 0.0000   | 15  | 0.94 |
| semantic alert  | 0.980 | 0.250 | 0.150 | <b>0.250</b> | -0.730   | 0.1986  | 0.1986   | 15  | 5.21 |
| grounding       | 0.500 | 0.050 | 0.850 | <b>0.050</b> | -0.450   | 0.0645  | 0.0000   | 15  | 0.94 |
| snapshot        | 0.985 | 0.550 | 0.050 | <b>0.550</b> | -0.435   | 0.2000  | 0.0000   | 15  | 1.00 |
| cumulative      | 0.925 | 0.850 | 0.150 | <b>0.850</b> | -0.075   | 0.2394  | 0.0845   | 15  | 1.00 |
| dedup           | 0.992 | 0.350 | 0.550 | <b>0.350</b> | -0.642   | 0.2132  | 0.0063   | 15  | 3.31 |
| state monitor   | 0.800 | 0.450 | 0.150 | <b>0.450</b> | -0.350   | 0.3062  | 0.0096   | 15  | 4.73 |
| narration       | 0.100 | 0.650 | 0.150 | <b>0.650</b> | +0.550   | 0.2045  | WITHHELD | 15  | 5.88 |
| sequential step | 0.010 | 0.150 | 0.020 | <b>0.150</b> | +0.140   | 0.3612  | WITHHELD | 15  | 3.76 |

We therefore test every selection by a *paired* bootstrap over videos: each draw resamples the frozen video ids with replacement, and every cell of a task is re-scored on the identical drawn set, so between-video variance — which dominates, since videos differ far more from one another than thresholds do on any one video — is removed. Each draw then replays the *actual selection rule*, tie-band included, rather than a bare argmax. Two statistics follow: the **re-selection rate**, the fraction of draws in which the shipped choice wins again, against a chance rate of  $1/k$  for a  $k$ -cell grid; and a confidence interval on the **difference** between the selected cell and its best rival.

Resampling is over videos, never over ticks or events. A 400 s video contributes  $\sim 400$  ticks whose  $p_{hit}$  values are strongly serially correlated through one shared KV cache; treating them as independent returns an interval several times too narrow.

## 5 Results

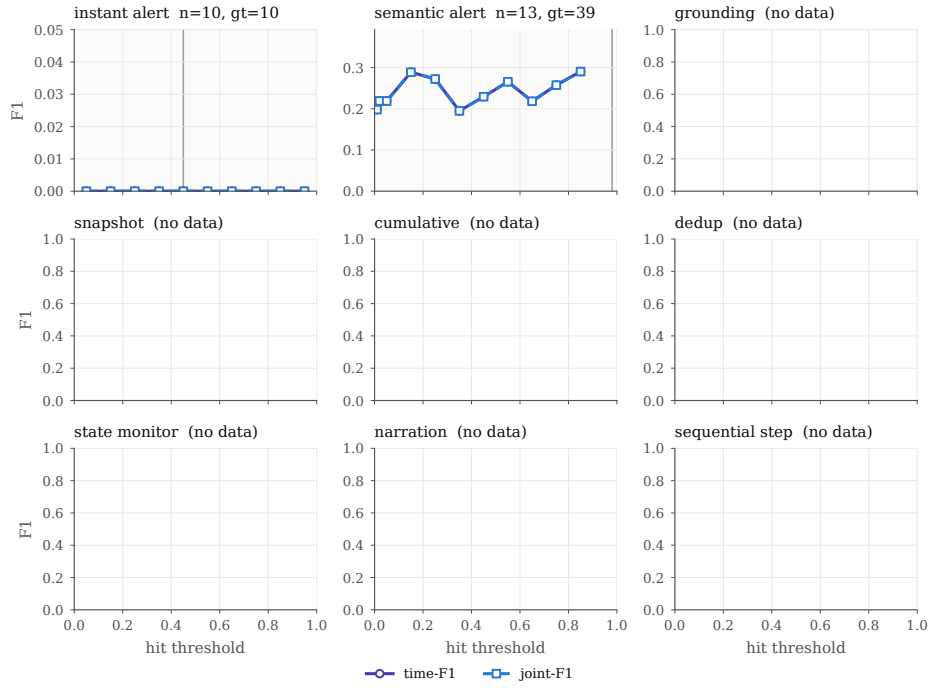
### 5.1 Pass 1: every task yields a winner

The pass-1 grid completed in full: 1,410 sample-evaluations over 94 cells, every cell scored on its complete 15-video set. The selection rule of Section 4 returned a threshold for all nine tasks (Table 1). Taken at face value this is a successful per-task fit, and Figure 1 shows the nine threshold curves it was read off.

### 5.2 The fits do not survive resampling

Table 1 reports, beside each selected threshold, the two statistics defined in Section 4. Across 2,000 paired bootstrap draws:

- **Every task’s  $\Delta$ time-F1 against its best rival has a 95 % interval containing zero.** No selected threshold is distinguishable from the best alternative in its own grid.



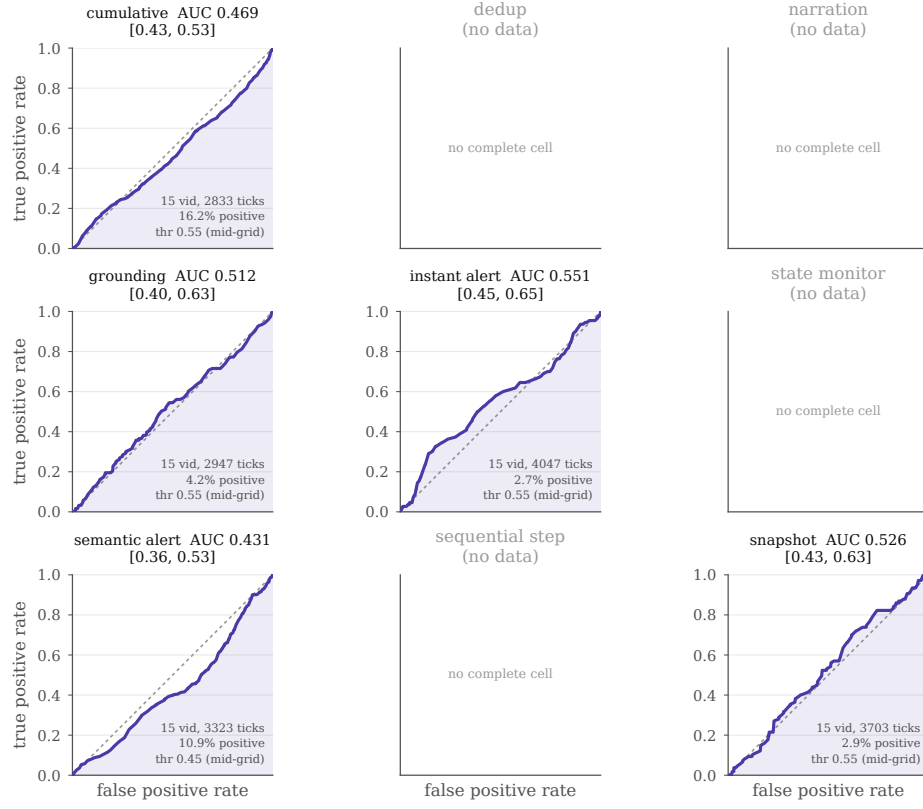
**Fig. 1.** F1 against gate threshold, one panel per task. Nine tasks are nine panels, not nine colours on one axis. Panel titles carry  $n$  and  $n_{gt}$ ; all panels share one  $y$  scale.

- **Re-selection rates are 25–61 %** against chance rates of 8–10 %. The curves are therefore not pure noise — the argmax is not uniform over the grid — but they are far from identifying a unique operating point.
- **semantic\_condition\_alert** re-selects at **6 %** against an 8 % chance rate: its fitted value reproduces *less* often than a cell drawn at random, and the bootstrap prefers 0.15 (33 %) over the selected 0.25.

Two entries in the threshold table are not fits at all, and are reported as such rather than carried into the final configuration:

*instant\_event\_alert* scores  $time-F1 = 0.000$  at all ten thresholds. Its apparent 100 % re-selection is a ten-way tie broken deterministically by the selection rule, not agreement between draws. The rule correctly flags the task FLAT and retains its shipped value. The system never places a single emission within  $\pm 3s$  of a ground-truth event on this task at any gate setting.

*explicit\_target\_grounding* alternates 0.065, 0.000, 0.061, 0.000 across the grid. With 15 emissions against 16 ground-truth events,  $F1 = 2 \text{ tp}/31$ , so those values are *one matched event, or zero*. The whole curve is a single coin flip per cell, and its interval is correspondingly the widest observed,  $[-0.207, +0.182]$ .



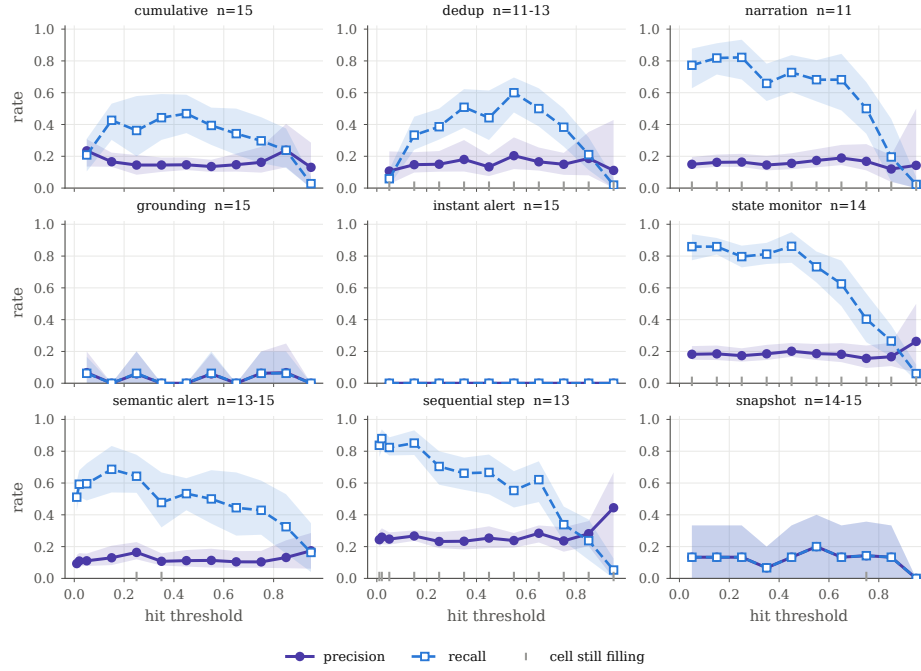
**Fig. 2.** ROC of per-tick  $p_{hit}$  against the  $\pm 3$ s positive label, with AUC and bootstrap 95% CI. AUC is threshold-free and therefore unaffected by this fitting: it measures the backbone, while F1 measures the gate.

### 5.3 Why: the gating score does not rank events

Figure 2 reports per-task ROC for  $p_{hit}$  against the  $\pm 3$ s positive label. AUC is 0.431–0.551 and every 95 % interval contains 0.5. Three innocent explanations are excluded:

- **Not a clock offset.** Sweeping the tick clock over  $\pm 10$ s in 2s steps leaves AUC within 0.39–0.55 with no peak anywhere; a genuine misalignment would produce one.
- **Not too tight a tolerance.** Re-running the label at  $\pm 10$ s leaves the result unchanged.
- **Not a degenerate score.**  $p_{hit}$  takes 232–284 distinct values spanning 0.000–0.997 with an interquartile range of 0.2–0.7. It is well spread; it simply does not track *when*.





**Fig. 3.** Per-sample precision and recall against threshold. The band is  $\pm 1$  s.d. clipped to  $[0, 1]$  and is a dispersion statistic, not a confidence interval: per-sample precision and recall are frequently exactly 0 or 1, so the distribution is not Normal. Actual per-sample values are overlaid.

`semantic_condition_alert` is below chance and falls monotonically as the offset goes positive (0.487 at  $-10$  s to 0.389 at  $+10$  s): its confidence is systematically *lower* near an event.

Figure 3 confirms this through an independent path. Per-sample precision is flat across the entire grid on every task (0.11–0.25 depending on task, with no trend), while recall falls monotonically. Firing less often buys no precision — which is what a ranking statistic at chance predicts, and which no threshold can repair.

#### 5.4 The gate has one identifiable degree of freedom

Table 2 pools all nine tasks. The fitted per-task arm scores highest in absolute terms (0.2544), but its paired advantage over a single global threshold is  $+0.0075$  with a 95 % interval of  $[-0.011, +0.027]$ . This is the *in-sample* comparison: the nine thresholds were fitted on these very videos and hold nine free parameters against the global arm’s one, and they still do not separate.

The global threshold, by contrast, is identified. Re-running the entire single-global sweep inside every bootstrap draw, 0.15 wins 94 % of them against a 10 %

**Table 2.** Ablation of the fit, pooled micro-averaged over 135 videos (fit subset). *global* 0.5 is the shipped gate’s effective behaviour (**hysteresis** tested  $p_{\text{hit}} \geq 0.5$  and ignored the per-task table); *shipped per-task* is the vision-only fit; *fitted per-task* is this work. <sup>†</sup>Threshold not on the run grid; served by the nearest cell actually evaluated (max. error 0.050). These rows are approximations. The single global optimum is stable: re-selected in 94% of 2000 paired bootstrap resamples of the videos (chance 10%). The per-task fits are not (25–61%; see text). joint-F1 is WITHHELD: the judge was unreachable for this run and content verdicts are never guessed.

| gate                          | time-P | time-R | time-F1       | $\Delta$ vs best global | [95% CI]                |
|-------------------------------|--------|--------|---------------|-------------------------|-------------------------|
| global 0.5 <sup>†</sup>       | 0.1421 | 0.5650 | 0.2271        | −0.020                  | [−0.039, −0.003]        |
| best single global (0.15)     | 0.1554 | 0.6013 | <b>0.2469</b> | —                       |                         |
| shipped per-task <sup>†</sup> | 0.1505 | 0.3561 | 0.2115        | −0.035                  | [−0.055, −0.016]        |
| fitted per-task               | 0.1628 | 0.5821 | 0.2544        | +0.007                  | [−0.011, +0.027] (n.s.) |

chance rate (next best 0.55, at 2%). The shipped per-task table is significantly *worse* than a flat 0.15 (−0.0354, interval excluding zero), driven by recall: 0.356 against 0.601. Since the shipped gate in fact fired at a fixed global 0.5, moving that single operating point is the one intervention in this study with a stable, reproducible effect.

## 5.5 Held-out evaluation

The claims above are measured on the 135-video fit subset and are therefore in-sample for the fitted arm. The complete 2,700-sample evaluation, the fit-disjoint rescore, and the per-task breakdown are [PENDING RUN]; Tables 5.5 and 5.5 are reserved for them. We record here, in advance of that run, the prediction that its pooled time-F1 under the fitted per-task thresholds will land within the 0.03 noise floor of the same run under the shipped thresholds. If that prediction fails, the fit is real and this section is wrong.

**Table 2 not yet available.** Requires stage 3 (OVERALL.json).

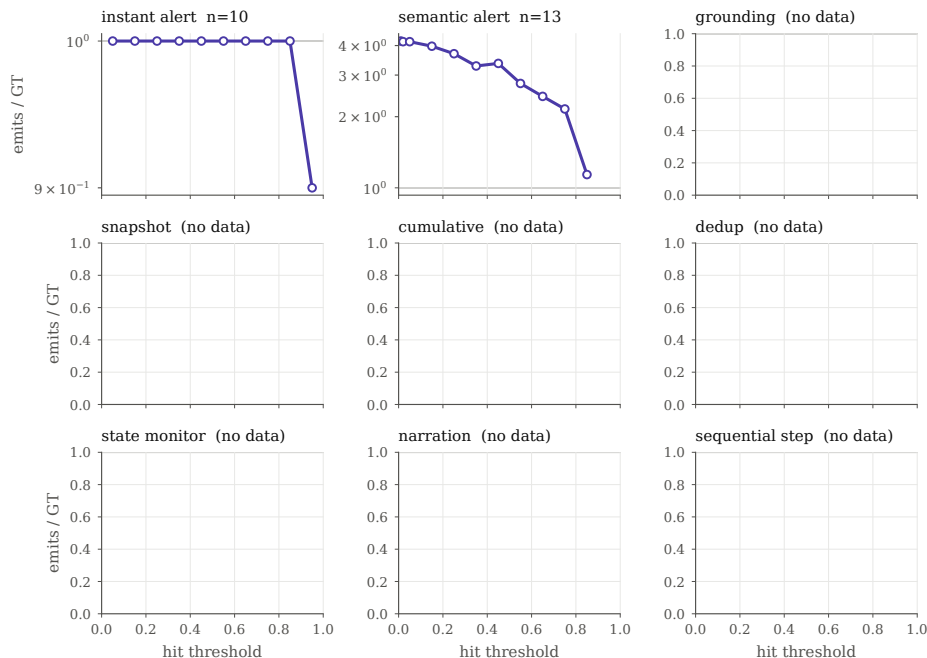
**Table 3 not yet available.** Requires stage 3 (OVERALL.json).

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## 6 Discussion

### 6.1 One degree of freedom, not nine

The three measurements in this paper are one mechanism observed three ways. AUC says  $p_{\text{hit}}$  cannot rank moments within a video. Per-sample precision being



**Fig. 4.** Emission calibration: emissions per ground-truth event against threshold, log axis, reference line at 1.

flat across the entire threshold grid, while recall falls monotonically, is the same fact expressed in the metric we optimise: firing less often buys no precision, which is exactly what a ranking statistic at chance predicts. And the selection audit says that what remains — a shallow, noisy dependence of F1 on emission volume — is not enough to identify nine separate operating points from fifteen videos each.

What makes the result more than a null is that the same quantity *is* identifiable when the data are pooled. The single global threshold re-selects at 94% against a 10% chance rate, while the per-task fits sit at 25–61%. Nothing about the tasks changed between those two numbers; only the number of videos backing each estimate did. This is a statistical power statement, not a claim that the tasks are secretly identical — their curves do differ, they are simply each estimated too noisily to rank.

## 6.2 Why a per-task table is the tempting wrong answer

Per-task thresholds are attractive because the tasks genuinely differ in emission rate, and because a per-task table always fits the fitting set better than a single value. Our fitted arm does score highest in Table 2. It is only when the comparison is made *paired*, against the best single global threshold, that the

advantage turns out to be  $+0.0075$  with an interval from  $-0.011$  to  $+0.027$  — and that comparison is in-sample, where nine free parameters are given every advantage over one and still fail to separate. A study that stopped at the four point estimates would have reported a per-task gain in good faith.

### 6.3 What would have to be true for per-task fitting to work

The obstruction is ranking quality, not grid resolution, and this distinction has a practical consequence: widening or refining the grid cannot help, because no threshold recovers information the score does not carry. Two things would change the answer. First, a gating score that actually ranks event-adjacent ticks above quiet ones — AUC meaningfully above 0.5 — would make precision respond to the threshold, and per-task optima would become separable. Second, more videos per task would shrink the per-task intervals; our own pooled result is the existence proof that the estimator works given enough data. The two are not independent: the number of videos needed scales with how weakly the score ranks.

### 6.4 Implications for how streaming gates are reported

We would draw a general methodological point from a specific negative result. A per-task threshold table is a set of fitted parameters, and it should be held to the standard applied to fitted parameters: a demonstration that the values recur under resampling, and a comparison against the simplest model that could explain the same data — here, one global threshold. Both tests are cheap, since they re-use the runs already performed and cost no additional GPU time. Neither is standard practice, and in this study each of them independently overturns the conclusion that the point estimates alone would have supported.

Finally, the two defects in Section 3 deserve emphasis beyond this system. The inert threshold would not have been visible in any aggregate metric: the system ran, emitted, and scored plausibly, and the fitted values were simply being ignored. It was found by checking that no fire had ever occurred below the global constant, which is a check on the mechanism rather than on the result. A grid search cannot detect that the parameter it is sweeping is disconnected.

## 7 Limitations

*The fit subset is small, and the negative results are bounded by it.* Thresholds are fitted on 15 videos per task. The AUC confidence intervals straddle 0.5 and the per-task  $\Delta F1$  intervals straddle zero; the claim these support is *indistinguishable from chance at this  $n$* , never *proven to be chance*. The same audits are re-run on the full 2,700-sample evaluation, where  $n = 300$  per task, and that is where the claim should be read at full strength.

*Two arms of the ablation are approximations.* `shipped_per_task` uses values such as 0.992 and 0.985 that no cell was ever run at, and 0.5 is not a grid point either — the grid steps  $0.45 \rightarrow 0.55$ , so the *global 0.5* row is in fact measured at 0.45. Each substitution and its distance is recorded, and these rows must be read as approximations. Only the *best single global* and *fitted per-task* rows are exact.

*The best single global threshold is itself an argmax.* We test it the same way we test the per-task fits — re-running the sweep inside every bootstrap draw — and it survives at a 94 % re-selection rate. That is evidence of stability on this subset, not a guarantee that 0.15 transfers; the held-out evaluation is the arbiter.

*Content metrics are withheld, not measured.* The judge was unreachable for these runs, so content-acc and joint-F1 are WITHHELD throughout. Selection ran on time-F1, a documented deviation from the pre-registered joint-F1. It is possible that a threshold which is indistinguishable in timing is separable in content quality; that possibility is untested here, and it is the single largest gap in this study.

*One backbone, one benchmark.* All results are for Qwen2.5-Omni-7B on OmniPro Online. The claim that a per-task threshold table is unidentifiable at this sample size is a claim about the interaction of this score’s ranking quality with this benchmark’s per-task sample count. A backbone whose  $p_{\text{hit}}$  ranked events well would plausibly support per-task fitting on the same data.

*The evaluation is not bit-reproducible.* Run-to-run variance on a fixed configuration reached time-F1 0.255 versus 0.051 on one task. Every comparison here is made against a 0.03 noise floor declared in advance, but individual cell values should not be read to more precision than that floor allows.

## 8 Conclusion

We set out to fit per-task emission thresholds for a proactive streaming assistant and report the gain. The fit completed, returned a threshold for every task, and would have supported that report. It does not survive being asked whether it recurs: no task’s selected threshold beats its own best rival outside noise, the selections re-appear in only 25–61 % of bootstrap resamples, and nine fitted thresholds together fail to beat a single global one even in-sample.

The mechanism is measured rather than inferred. The gating score does not rank event-adjacent moments above quiet ones — AUC 0.431–0.551, every interval covering chance, robust to clock offset, tolerance and degeneracy checks — so a threshold on it can only purchase emission volume. Volume has one useful setting, and pooled over 135 videos that setting is recovered stably, at a 94 % re-selection rate, while the same quantity split nine ways is not recoverable at all.

The practical recommendation for this system is therefore a single global operating point rather than a per-task table, and the shipped per-task values — inherited from a vision-only backbone — are measurably worse than a flat threshold. The methodological recommendation is more general: a per-task threshold table is a set of fitted parameters and should be reported as one, with a resampling check that the values recur and a comparison against the simplest model that explains the same data. Both tests re-use runs already performed and cost no additional compute. In this study each of them, independently, reverses the conclusion the point estimates alone would have supported.